

Attack Navigation Layer Coverage Map

Project

SecOpsAI – AI-Powered Behavioral Threat Detection and Adversarially Robust Security Operations Platform

Purpose

The Attack Navigation Layer provides a high-level view of potential attack paths within the SecOpsAI platform. It identifies critical assets, possible attacker actions, affected components, and security controls implemented to detect, prevent, or mitigate threats.

Attack Surface Overview

The following components form the primary attack surface of the SecOpsAI platform:

- API Layer
- Alert Pipeline
- Data Pipeline
- ML Engine
- Dashboards
- Documentation and Reporting Services
- Configuration Files
- Authentication and Authorization Services

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Attack Navigation Coverage Map

Entry Point	Target Component	Attack Technique	Potential Impact	Security Controls
Public API	API Services	Unauthorized Access	Data Exposure	Authentication , Authorization
Public API	Data Pipeline	Injection Attacks	Data Corruption	Input Validation
User Dashboard	Dashboard Services	Privilege Escalation	Unauthorized Actions	RBAC
Data Sources	ML Engine	Data Poisoning	Incorrect Detection Results	Data Validation
Alert Pipeline	Alert Management	Alert Manipulation	Missed Security Events	Integrity Verification
Configuration Files	Systems Services	Misconfiguration Exploitation	Service Compromise	Configuration Reviews
Credentials	Authentication Services	Credential Theft	Account Takeover	MFA
Report and Documentation	Information Assets	Information Disclosure	Sensitive Data Exposure	Access Control

Attack Flow Examples

Scenario 1: Unauthorized API Access

Attacker targets exposed API endpoint.

Attempts credential guessing or token abuse.

Gains unauthorized access to system resources.

Security controls detect abnormal access patterns.

Access is restricted and alerts are generated.

Scenario 2: Data Poisoning Attack

Malicious data enters the ingestion pipeline.

Data reaches the ML Engine.

Model behavior becomes unreliable.

Validation mechanisms identify anomalies.

Suspicious datasets are quarantined for review.

Scenario 3: Dashboard Privilege Escalation

Attacker compromises a low-privilege account.

Attempts privilege escalation.

RBAC policies prevent unauthorized access.

Security logs capture the activity.

Security team investigates generated alerts.

Security Monitoring Coverage

The platform provides monitoring for:

Authentication Events

Authorization Failures

API Abuse Attempts

Data Integrity Violations

Alert Manipulation Attempts

ML Model Anomalies

Privilege Escalation Attempts

Configuration Changes

Conclusion

The Attack Navigation Layer documents potential attacker movement throughout the SecOpsAI environment and maps available defensive controls. This coverage map supports threat modeling, risk assessment, detection engineering, and continuous security improvement activities.